

COM-480 DATA VISUALIZATION
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Process Book

The Price of Wind and Sun

How Germany's renewable build-out reshapes wholesale electricity prices across Switzerland, France, Italy and Austria.

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com-480-data-visualization.github.io/HSquareB

Where we started

1.1 • THE ORIGINAL IDEA

We started with a question that felt simple on the surface: how does the build-out of renewable energy in one country affect electricity prices in its neighbors?

The dataset gave us the raw material to answer it. We used the **ENTSO-E Transparency Platform** download covering 1 January 2024 through 30 June 2025: 301,391 hourly observations across 23 bidding zones, with day-ahead prices and generation disaggregated by fuel type. We filtered to five zones sharing physical interconnectors across the Alps and the Rhine: Switzerland (CH), Germany-Luxembourg (DE_LU), France (FR), Northern Italy (IT_NORD), and Austria (AT).

Our first plan was clean: a choropleth of the five countries, a slider to move through time, and a toggle between two colour modes (price on one, renewable output on the other) so the reader could trace the relationship for themselves.

1.2 • WHY THAT WAS NOT ENOUGH

Two things happened when we started working seriously with the data and thinking about who would use our website.

1

The EDA kept surprising us

The numbers kept arriving in ways that a two-mode toggle would have buried. Each finding pointed in a different direction, and leaving the reader to discover all of them by switching colour modes assumed both domain knowledge and patience that most readers would not have.

2

The audience did not speak the language

The second was the audience problem. Electricity markets have their own vocabulary (bidding zones, merit-order, duck curve, Dunkelflaute), and a reader without it, handed two colour modes, would see the correlations without understanding why any of them were remarkable. Showing the relationship is not the same as explaining why it matters.

Those two realisations, together, pushed us toward a fundamentally different structure.

What the data showed us

Before settling on any design, we spent time understanding what the data actually contained. The EDA findings drove every significant design decision that followed: what to show first, what to suppress, and what to hand over to the reader.

The Italy divide

Over 18 months, Germany recorded 846 hours of negative prices (6.5% of all hours), France 715 (5.5%), Austria 584 (4.5%), Switzerland 529 (4.0%). Northern Italy: zero. Not a handful, but zero across the entire window. Italy sits behind the Alps, relies heavily on gas-fired generation, and is two interconnectors further from Germany's solar core. That divide became the narrative's counterpoint: Italy as the insulated market that makes Switzerland's 529 hours feel structural rather than accidental.

Switzerland should not be doing this

Switzerland has almost no utility-scale solar and wind, and a generation mix dominated by flexible hydro. Nothing in its own fundamentals should push its clearing price below zero. Yet 529 times it did, and 88.8% of those hours fell inside the same clock hour as a German negative. Austria sat at 89.7%, France at 80.6% (France's higher absolute count includes domestically produced negatives from its nuclear-plus-wind stack; Switzerland has no such stack). Switzerland is not producing those negative hours. It is importing them through the interconnectors.

Five mixes, five price identities

Germany peaks near 70% renewable in summer (solar); Austria near 84% year-round (hydro); France a flat 20-30% (nuclear). Each mix produces a distinct price character, and the five together explain why the same solar event lands so differently across the region.

The duck curve is deepening fast

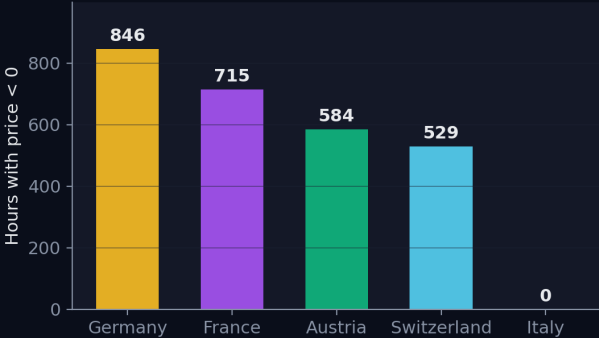
Germany's average hour-13 price fell from +€16.17/MWh in May 2024 to -€12.15/MWh in May 2025. On a *monthly average* basis, the belly of the duck crossed below zero inside a single year. This is not an extreme-event statistic. It is a structural shift in how the market clears at the most common demand hour.

529
CH NEG. HOURS

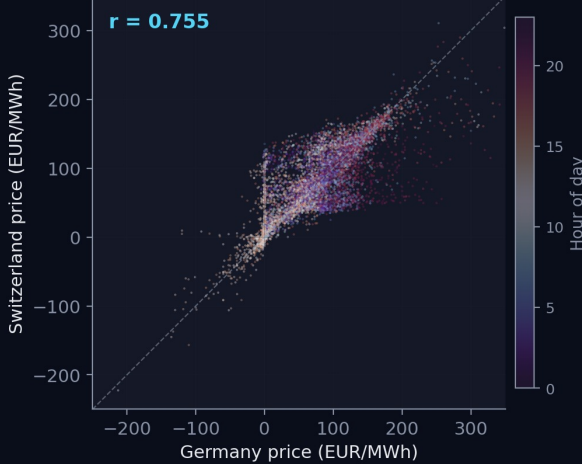
88.8%
CO-OCCUR DE
ITALY NEG.

r=0.755
CH-DE CORR.

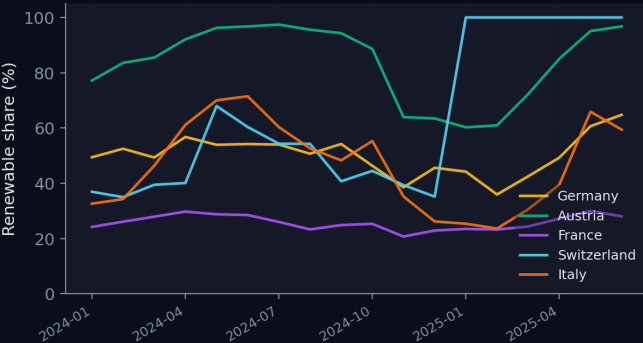
Negative-price hours by country, Jan 2024 - Jun 2025



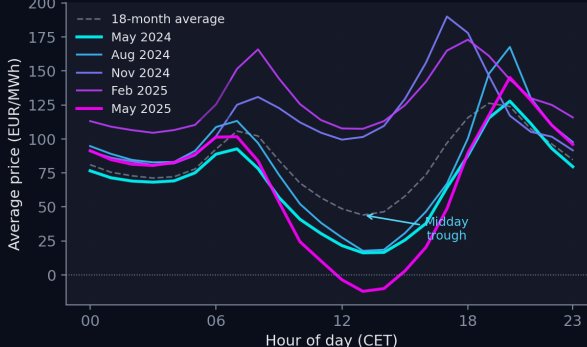
Hourly prices: Switzerland vs Germany



Monthly renewable generation share by country



Germany's duck curve is deepening



Telling the story first

The EDA left us with findings that were individually striking and collectively coherent, but they needed to be narrated, not just displayed. A reader who arrives at a choropleth map without context will see colours change but will not know what they mean.

We decided to show each key finding directly, in sequence, with enough framing that a reader without domain knowledge could follow the argument. That decision produced the scrollytelling architecture: a sticky map beneath a column of seven editorial cards, each card sending a state change to the map (hour, colour intensity, flow arrows, annotation). The map never leaves the screen during the scroll.

3.1 • THE SEVEN STEPS

Hook Key visual

- 1 **Midnight baseline** Map at 00:00
- 2 **Solar ascending** Map at 11:00; Germany wakes up; solar donut
- 3 **The shock** 13:00; -€145.12; flow arrows
- 4 **The pattern** Calendar heatmap; DE/CH toggle
- 5 **The mechanism** Germany's generation stack
- 6 **The duck deepens** Animated monthly profiles
- 7 **Five identities** Small-multiples grid of annual 24-h profiles

ONE REVISION THAT MATTERED

The original sketch showed Step 3 as a full-day price line chart. The line chart required the reader to scan a curve and locate the minimum, whereas the single number **-€145.12** at typographic scale landed immediately. So we demoted the line chart (keeping it at the bottom of the card as a small supporting element rather than the focus), promoted the number to a full-card bloom, and gave Germany's simultaneous price (-€135.45) in a sentence alongside it for scale.

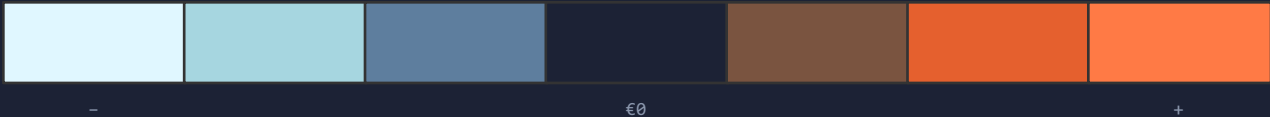


The seven steps run from the most evident to the most subtle so a reader without domain knowledge is taught the vocabulary in the order they need it.

The visual language

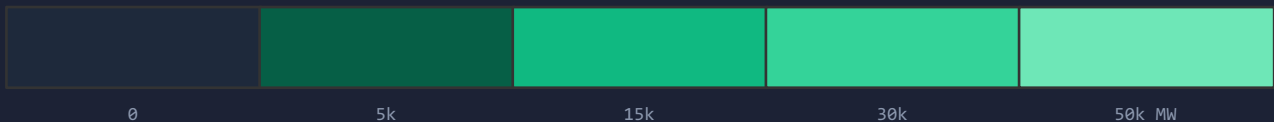
Energy data is often shown in dense technical dashboards. We wanted the site to read like long-form journalism, so every visual choice ties back to a small, consistent system.

PRICE COLOUR SCALE



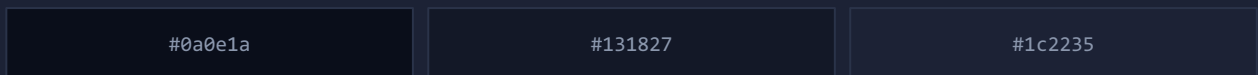
Diverging from ice-white at deep negatives, through navy at zero, to orange-red at peaks. The inversion (brightest for most negative) is deliberate. Negative prices are the drama, and they should carry the most visual weight on a dark background. A conventional blue-cold scale would have made Switzerland's lowest hour look merely "cold"; this scale makes it glow.

RENEWABLE COLOUR SCALE • 0 → 50,000 MW



A dark-to-bright green ramp for renewable generation. Near-black navy means almost no renewable output; mint at the top means peak generation (Germany at full solar). Green keeps the renewable signal semantically separate from the price scale, so the two never compete in the reader's eye.

BACKGROUND RAMP



The site is dark-only. A three-level navy ramp gives sections quiet hierarchy without changing perceived brightness. Colour is reserved for data; the canvas itself stays out of the way.

TYPOGRAPHY

Fraunces

Variable serif for editorial copy and hero numbers.

JetBrains Mono

For every data value, label, country code, and legend tick.

Numbers are never set in the serif.

The pairing gives the body editorial warmth and the data machine precision.

FLOW ARROWS



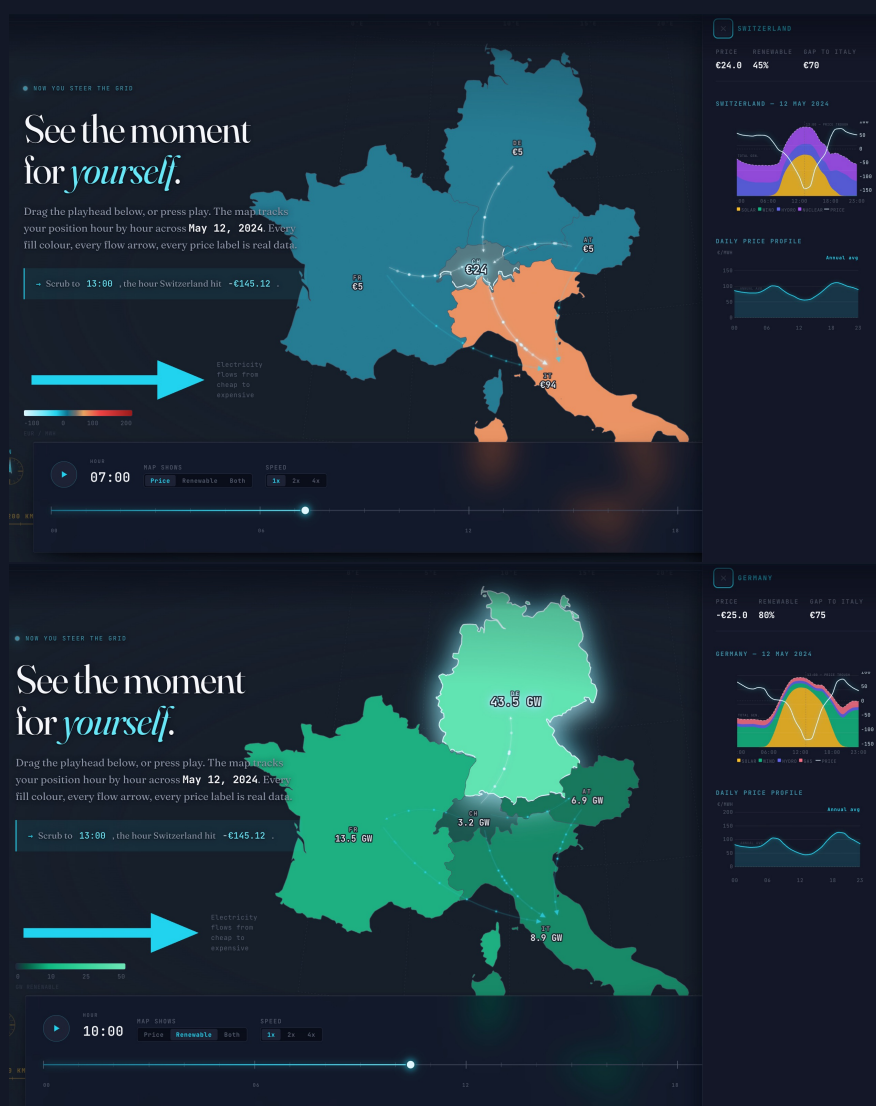
Width is proportional to the price spread between zones; direction is inferred from the differential (electricity flows from cheap to expensive).

Drawn as quadratic bezier curves, control points offset 18% to the left so adjacent flows never overlap; endpoints inset 28 px to clear the labels.

Letting the audience explore

Ending at the final narrative card would have closed something that should stay open. The patterns we found during EDA were the ones we already knew to look for. A reader coming to the data fresh might notice something we missed entirely: a country pair we did not highlight, an hour of the day we did not linger on, an anomaly in the months between our showcase events.

So after Step 7, the page opens into the **interactive explorer**. The same map, now driven by a 24-hour timeline scrubber over 12 May 2024. The reader can play the day forward at 1x, 2x, or 4x speed, pause with the Space bar, or step frame by frame with the arrow keys. Clicking any country opens a sidebar showing the current price, renewable share, gap to Italy, a mini generation stack, and the daily price profile with the annual average overlaid as a ghost line.



COLOUR-MODE TOGGLE

We provide a colour-mode toggle that gives the readers three views of the same map:

Price mode

Colour countries by day-ahead clearing price colours.

Renewable mode

Colour countries by total renewable generation in GW.

Both mode

Overlays the two other modes simultaneously: country fills stay on the price scale while pseudo-3D bars rise beside each label, their height encoding renewable output, so the reader can see Germany's solar peak and the price collapse side by side at the same hour.

The Both mode was the last major feature added since neither price nor renewable alone told the causal story as clearly as the two together.

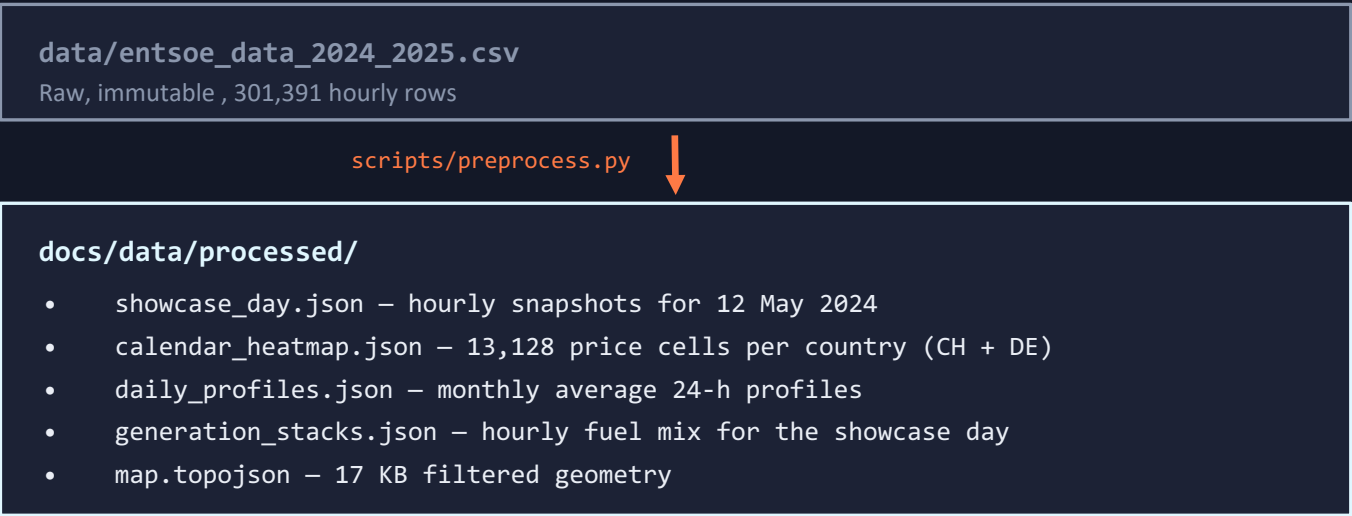
Building it

THE STACK

Rendering	D3 v7 — SVG + Canvas, vanilla ES modules
Scroll	Scrollama v3
Map	world-atlas TopoJSON, 5 countries (17 KB)
Preprocessing	Python 3.9, pandas, pyarrow
Hosting	GitHub Pages, static from docs/

No framework, no build step. Every ES import is resolved from a browser importmap. Avoiding a bundler kept the dev loop instant (edit, then refresh) and removed an entire class of dependency-resolution bugs.

DATA PIPELINE



showcase_day and *daily_profiles* preload at parse time; the 182 KB heatmap is deferred until Step 4 so it never blocks first paint.

340 ms → 4 ms

The calendar heatmap uses a Canvas renderer rather than SVG. The first SVG implementation took about 340 ms to render and caused layout thrash each time the DE/CH tab toggle fired. Moving to Canvas brought repaint time to about 4 ms via `ctx.clearRect` plus synchronous redraw. The Canvas element carries an *aria-description* so screen readers still receive the data summary.

MOBILE

Making the desktop-only build mobile-friendly required responsive layouts (collapsing grids below 600px, full-screen sidebars, flexible heatmap cells), touch optimizations (44px targets, suppressed hovers), iOS safe-area insets, and a custom resize handler to recompute modules on orientation change.

Decisions & peer assessment

6 • DECISIONS THAT WERE NOT OBVIOUS IN ADVANCE

REFRAME

Turning the Italy problem into an asset

We originally framed Italy as a full participant in the negative-price story. When the EDA showed zero negative hours, that felt like a gap. We reframed it: Italy's insulation is the counterpoint that makes Switzerland's 529 hours feel structurally surprising rather than incidental.

SIMPLIFY

Hero number, not a line chart

Step 3 went through two iterations. A full-day price line chart asks the reader to scan and locate the minimum. The number –€145.12 at typographic scale hits in under a second. We chose the number and moved the line chart to the explorer sidebar, where the reader encounters it only after they have already absorbed the peak-hour story.

ADD

A third 'Both' mode

The original explorer offered two colour modes: Price or Renewable. Toggling between them made the correlation between solar output and price collapse visible but effortful, because the reader had to hold one map in memory while looking at the other. We added a third mode, Both, that keeps the price colour scale on the country fills while pseudo-3D bars rise beside each label to show renewable output in GW. A square-root height scale ensures smaller producers like Switzerland and Austria remain legible next to Germany's much larger output. Two legends appear simultaneously, one for each encoding.

COLOUR

Ice-white for the most negative prices

We tested three colour scales before settling on the current one. A red-to-blue diverging scale made negative prices look "cold", which is semantically misleading since they represent economic stress. A green-negative, red-positive scale violated traffic-light conventions. The ice-white to navy to orange scale was the only one where Switzerland going lighter than Germany at Step 3 was immediately visible on the dark background without any annotation pointing to it.

7 • PEER ASSESSMENT

All three of us contributed across every stage of the project. The framing of the question, the choice of dataset, the move from a single choropleth to a guided narrative followed by an explorer, the seven-step arc, the visual language, and the final round of edits were decisions we made together through repeated team discussion. The notes below describe where each of us focused, but every part of the piece carries traces of all three.

Brian Banna

SCIPER 356437

Built the Python preprocessing pipeline that converts the raw ENTSO-E CSV into the JSON the site uses, the D3 map module with its flow-arrow layer and live price labels, the seven-step scroll orchestration on Scrollama, and the interactive explorer with its timeline scrubber and country sidebar. He put together the visual design system and ran the mobile and accessibility passes.

Thao Huyen Le

SCIPER 355566

Shaped the seven-step narrative arc and wrote the step copy, including the Step 3 hero framing and the Italy closer. After Milestone 2, added the Both mode to the explorer, pseudo-3D bars rising beside each country to show renewable output alongside the price colouring. Wrote this process book.

Hajj Hannah

SCIPER 346545

Ran the early exploratory data analysis that produced the findings the piece is built around (the negative-hour distribution across the five zones, the Switzerland-to-Germany co-occurrence rate, and the year-over-year duck-curve shift). Helped compose the Step 7 small multiples.